

Lasso Regression Interview Questions & Answers

For Data Scientist, Junior ML Engineer, and Data Analyst interviews.

Q1. What is Lasso Regression?

Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a linear regression technique with L1 regularization that improves model generalization by adding a penalty. This penalty not only controls overfitting but also performs automatic feature selection by shrinking some coefficients exactly to zero, making Lasso useful in high-dimensional datasets where interpretability and sparsity are important. Aggressive shrinkage

- Uses L1 regularization to reduce overfitting
- Encourages sparse models (zero coefficients)
- Automatically performs feature selection
- Useful for high-dimensional datasets

Note: L1 regularization adds the sum of absolute values of model weights to the loss function, encouraging sparsity by pushing some weights exactly to zero.

Q2. Why do we need Lasso?

In our dataset, some features may be: irrelevant, redundant, noisy, highly correlated with other features. An ordinary Linear Regression doesn't have a mechanism specifically designed to remove them. This is where regularization comes in.

Q3. Ridge vs Lasso: the big idea?

Both methods start with the ordinary Linear Regression loss: $RSS = \sum (y_i - \hat{y}_i)^2$
Then they add a penalty for large coefficients.

Ridge: Ridge adds the sum of the square of the coefficients to the loss function

Lasso: adds the sum of absolute values of model weights(coefficients) to the loss function
The important difference is that **Ridge uses L2. Lasso uses L1.**

Q4. Why is it called "sparse"?

A sparse vector contains lots of zeros.

For example: $[5.2, 0, -1.3, 0, 0, 2.7]$ $[5.2, \backslash; 0, \backslash; -1.3, \backslash; 0, \backslash; 0, \backslash; 2.7]$

There are several zero coefficients.

Therefore:

Lasso $\rightarrow$ many coefficients become 0 $\rightarrow$ sparse coefficient vector $\rightarrow$ sparse model

This is one of the most important things to remember.

Q5. WHY does Lasso produce zero?

Look at the two penalties.

- **Ridge:** β^2 . If we visualise it. It's a smooth curve. Ridge is primarily a coefficient shrinkage technique. It pulls coefficients toward zero. But can't make them zero. So Ridge usually keeps all features.

Ridge $\rightarrow$ reduce

- **Lasso:** $|\beta|$. If we visualise it. It produce sharp corner around the curve. The L1 penalty creates a situation where the optimization can settle exactly at zero.

Lasso performs coefficient shrinkage + feature selection.

Lasso → reduce + eliminate

Q6. What does λ do?

λ controls how strongly we penalise the coefficients.

Small λ :

- Penalty is weak.
- Few or no coefficients become zero.
- The model focus more on the training data. The model fits well on the training data but failed to fit on the testing data.
- Low bias and high variance. Risk of overfitting.

Large λ :

- Penalty become strong.
- More coefficients may become zero.
- Increasing λ generally increases regularization and can produce a sparser model.
- High bias and low variance.
- Better generalization
- But be careful: **Very large λ can cause underfitting.**

Q7. How Lasso balances this Bias-Variance tradeoff?

- Penalization reduces variance by simplifying the model
- Feature elimination may increase bias slightly
- Net effect is often lower test error

Q8. What happens when $\lambda = 0$? When $\lambda = 0$ then the coefficients become zero and no regularization will happen and Lasso becomes ordinary Linear Regression.

So when $\lambda = 0 \rightarrow$ Linear Regression

As λ increases: More regularization $\rightarrow$ coefficients shrink $\rightarrow$ some can become zero

Q9. What does "embedded feature selection" mean?

There are different approaches to feature selection.

1. **Filter methods:** You select features before training the model. For example:

- correlation
- chi-square
- mutual information

2. **Wrapper methods:** You repeatedly train models and evaluate subsets. For example:

- Recursive Feature Elimination (RFE)

3. **Embedded methods:** Feature selection happens during model training. Lasso is an embedded feature-selection method because: $\beta_j = 0$

means that feature is effectively removed from the model.

"So is Lasso a feature selection method?" A strong answer is:

"Yes. More specifically, Lasso performs embedded feature selection by shrinking some coefficients exactly to zero."

Q10. Does Lasso needs feature scaling? Should we scale features before Lasso?

Lasso penalizes coefficients. Therefore, the scale of the features matters. If features are on very different scales, the penalty can affect them unfairly. That's why we generally standardize features before applying Lasso. Then the features are on a comparable scale. This is also true for Ridge.

"Yes. Since Lasso penalizes the magnitude of coefficients, feature scaling is important so that the regularization treats features fairly.

Q11. What happens with correlated features?

Suppose we have:

Feature A = house size

Feature B = number of rooms

These might be highly correlated. Lasso can sometimes select **one feature and set the other to zero**.

For example:

$$\beta_A = 2.4 \quad \beta_B = 0$$

even though both contain useful information.

This is one reason Ridge can sometimes be preferable when we have many correlated predictors.

Ridge tends to distribute the coefficients across correlated features rather than aggressively selecting one.

Q12. Lasso vs Ridge vs Elastic Net?

	Linear Regression	Ridge	Lasso	Elastic Net
Regularization		L2	L1	L1 + L2
Shrinks coefficients				
Can make coefficients 0		Generally		
Feature selection				
Handles correlated features	—	Good	Can be unstable	Often better
Main idea	Fit data	Shrink	Shrink + select	Shrink + select

Note: Elastic Net combines: L1+L2. So it tries to get the benefits of both.

Q13. Applications of Lasso Regression.

- **Feature Selection in ML:** Removes irrelevant predictors automatically
- **Genomics:** Selects influential genes from thousands of variables
- **Finance:** Identifies key risk factors and predictors
- **Text Mining:** Chooses important words in high-dimensional text data
- **Econometrics:** Builds compact explanatory models

Q14. Advantages of Lasso Regression

- **Automatic Feature Selection:** No extra preprocessing needed
- **Sparse Models:** Easier interpretation
- **Overfitting Reduction:** Improves generalization
- **High-Dimensional Suitability:** Works well when features are many
- **Computational Efficiency:** Fast optimization methods

Q15. Limitations of Lasso Regression

- **Correlated Feature Issue:** Selects one feature from a group arbitrarily
- **Bias Introduction:** Strong regularization may underfit
- **Hyperparameter Sensitivity:** λ must be tuned carefully
- **Limited Group Selection:** Not ideal for grouped predictors
- **Linear Assumption:** Cannot model complex non-linear patterns

Quick Interview Questions/Answers

Basic Questions

Q1. What is Lasso Regression?

Linear Regression + L1 regularization.

Q2. Why do we use Lasso?

To reduce overfitting and perform feature selection by shrinking some coefficients to exactly zero.

Q3. What is the Lasso objective function?

$RSS + \lambda \sum |\beta_j|$

Q4. What does λ control?

The strength of regularization.

Q5. What happens when λ increases?

Coefficients are generally shrunk more strongly and more coefficients may become zero, potentially producing a sparser model.

Intermediate

Q6. Why does Lasso create sparsity?

Because the L1 penalty $|\beta|$ has a sharp corner at zero, allowing the optimal coefficient to become exactly zero.

Q7. Why doesn't Ridge normally create sparsity?

Ridge uses a smooth L2 penalty β^2 , which shrinks coefficients toward zero but generally doesn't make them exactly zero.

Q8. Is Lasso feature selection?

Yes. It is an embedded feature-selection method.

Q9. Why should we standardize features before Lasso?

Because the penalty depends on coefficient magnitude, so features should be on comparable scales.

Advanced

Q10. What happens when predictors are highly correlated?

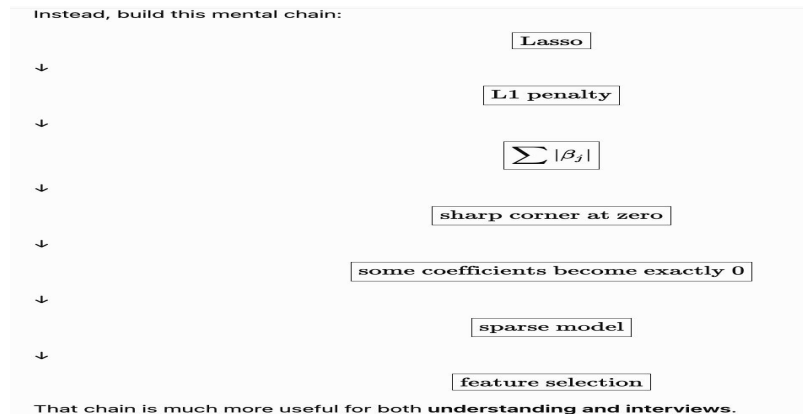
Lasso may arbitrarily select one correlated feature and shrink others to zero, whereas Ridge tends to distribute weights among correlated predictors.

Q11. What happens when λ is extremely large?

Coefficients can be heavily shrunk, potentially causing underfitting. In the extreme, coefficients can approach zero.

Q12. How do you choose λ ?

Usually through cross-validation, such as LassoCV, selecting the value that gives the best validation performance.



A final interview-level answer

Lasso Regression is Linear Regression with L1 regularization. It adds a penalty based on the absolute values of the coefficients to the loss function. The penalty discourages large coefficients and, because the L1 penalty has a sharp corner at zero, it can shrink some coefficients exactly to zero. Therefore, Lasso not only helps reduce overfitting but also performs embedded feature selection and produces a sparse model. The strength of this regularization is controlled by lambda, which is usually selected using cross-validation such as LassoCV.